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FSMB Embedded Systems Project Report

Project: Arduino-Based Voltage Protector

For this assessment, I built a microcontroller-based voltage protector prototype. The core idea is simple. To monitor the power line, and if things go out of bounds (overvoltage, undervoltage, or overcurrent), trip a relay to disconnect the load before it gets damaged. To ensure The device must feature computer communication capabilities as well as users must be able to change its settings either directly from the device itself or via a computer, I added a serial interface and physical buttons to let the user adjust the limits and set command.

How it works:
The brain of the system is an Arduino Uno (ATmega328P). In my Proteus simulation, it reads analog inputs representing voltage and current. It constantly compares these readings against configurable thresholds. If the values look good, the relay stays engaged and the load stays powered. If a fault is detected, the Arduino immediately de-energizes the relay to cut power to the appliance.

Communication & Controls:
I used UART for computer communication. Besides, I made sure to cross the TX/RX lines and share a common ground. The firmware accepts specific serial commands to update the threshold values. Locally, four push buttons (UP, DOWN, SET, RUN) let you navigate and change settings directly on the device without needing a PC. Over-Voltage, Under-Voltage, Over-Current Limit could be set by user besides it’s mode could be automated or manual through command.

Testing & Honest Limitations:
I tested the simulation under normal conditions, overvoltage, undervoltage, and overcurrent scenarios. I also verified that the serial commands and push buttons successfully update the thresholds. However, I want to be completely transparent that **this is a simulation-based result.** It does not prove real-world sensor accuracy, calibration, or electrical isolation. Because I am not using actual mains voltage, the relay behavior and sensor readings are idealized. A physical deployment would absolutely require proper isolation components and a fail-safe analysis. Any feature I couldn't fully verify in the simulation, I've marked as incomplete in my final assessment.

Safety Note:
As per the assessment guidelines, I did not build any hardware connected to mains voltage. Everything was simulated using low-voltage stand-ins in Proteus to emulate grid conditions safely.